

- 1 (a) Define acceleration.

.....
 [1]

- (b) In an experiment, two objects A and B are released from the side of a building, as shown in Fig. 1.1.

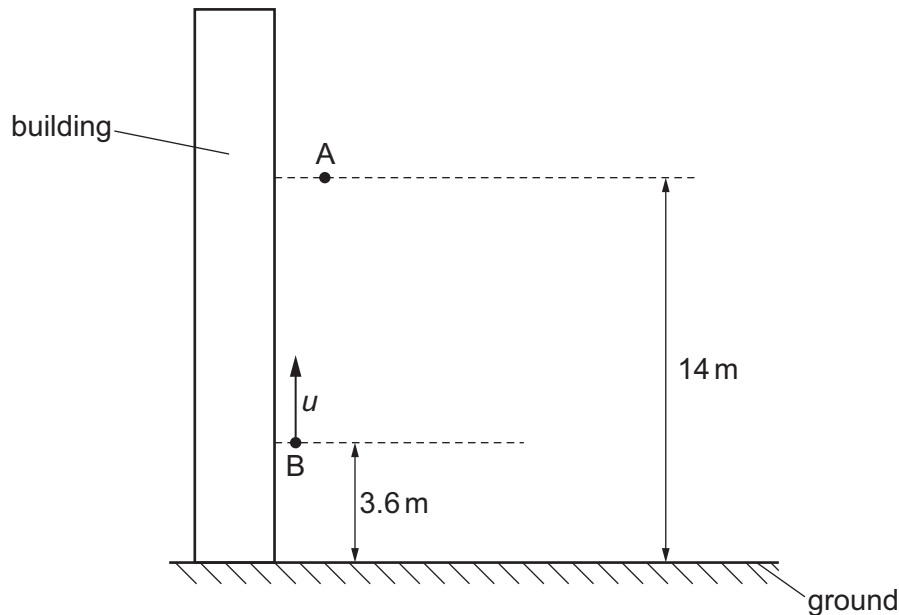


Fig. 1.1 (not to scale)

Object A is released from rest at a height of 14 m above the horizontal ground.
 Object B is released with an initial upwards vertical velocity u at a height of 3.6 m above the ground.
 Both objects take the same time to reach the ground and they do not collide with each other.
 Air resistance is negligible.

- (i) Calculate the time taken for object A to reach the ground.

time = s [2]

- (ii) Use your answer in (b)(i) to calculate u .

u = ms^{-1} [2]

